



Amrita School of Computing, Amritapuri Campus
Department of Computer Science and Engineering

23CSE498

Project Phase II

2026

B.Tech CSE — Initial Project Idea Submission

Project Team

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Guide Details

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1. Project Title

- Keep the title concise (8–15 words).
- The title may evolve later as the project becomes clearer.

Project Title:

Retrieval Beyond Text: Designing a Context-Aware Retrieval Framework for Multi-Party, Time-Dependent Conversations

2. Project Abstract

2.1 Motivation

- Explain the real-world problem.
- Identify the stakeholders or beneficiaries.
- Mention why the problem is important.
- Support with facts, observations, or statistics wherever possible.
- Keep it within 150–250 words.

Motivation:

WhatsApp is now the dominant medium for personal communication , now with more than 3 billion users exchanging over 150 billion messages every day, and India alone saw a increasing growth of over 341% in usage between 2021 and 2025. Now as these conversations piles up over years and years of group chats , personal discussions, plans or even informal exchanges it becomes increasingly difficult for users to recall or locate specific information buried in this history like old addresses, commitments etc. Now unlike documents this information is not organised , searchable or structured in any meaningful way , with whatsapp search only matches exact keywords that fail on paraphrased or vaguely remembered queries.

The stakeholders here are the users managing their personal communications and also small business owners that depend on WhatsApp as their primary coordination tool. Now for these users losing track of information in a chat history will be a major inconvenience that could lead to missed commitments and wasted time spend scrolling messages.

The reason why we chose this because it tackles two urgent needs that are , making a massive growing volume of everyday data useful while also not compromising user privacy . A retrieval augmented approach addresses this directly , Instead of exposing the entire history it fetches only the specific and relevant data while keeping the rest of the data untouched and private . This makes RAG a natural fit for building an assistant that can accurately and privately retrieve and reason over personal chat data

2.2 Problem Statement

- Clearly define the specific problem to be addressed.
- Mention the limitations of current practices or existing solutions.

- Ensure the problem is feasible to solve within the project duration.
- Avoid discussing implementation details or algorithms.

Problem Statement:

WhatsApp users generate large volumes of personal data everyday but this data remains difficult to search or make use of simple keywords lookups. Now WhatsApp's search feature only matches the exact text and so it fails when a user recalls a gist of a message but not the exact words.

So it offers no way to ask questions like "What did we decide about the trip?" or "Who was bringing the documents" etc. So as a result useful information may get buried in months or years of chat history and it becomes increasingly difficult to retrieve it later on and force us to manually scroll and re-read conversations to recall what happened.

Now existing AI assistants and retrieval solutions do not address this problem well. General purpose RAG frameworks are built to be evaluated on structured documents or a AI's own history with a user, but not on mass exported, multi party human conversations with fragmented messages, mixed languages and time sensitive content. Sending such content to a cloud based tool or a third party raises privacy concerns that most of the existing solutions don't adequately address.

So we addressed this problem of enabling users to retrieve relevant information from their own messages in a way that is accurate, context aware and privacy conscious.

3. Objectives

- Provide 3–5 clear and measurable objectives.
- Start each objective with an action verb such as Design, Develop, Analyze, Evaluate, Detect, Predict, Classify, etc.
- Objectives should directly support the problem statement.

Objectives:

1. Analyze the limitations of the existing RAG frameworks like LangChain, LlamaIndex, Haystack and memory architectures when used on fragmented, multi-party, time-dependent data such as WhatsApp chat exports.
2. Design a retrieval approach that works on this conversation data, which makes use of thread-aware chunking, speaker-aware context, and time-weighted ranking to address the specific limitations identified.
3. Develop a pipeline to process the exported WhatsApp chat data which include parsing, chunking, and embed the data to be suitable for natural-language querying.
4. Evaluate the proposed retrieval approach against baseline RAG frameworks using defined metrics like retrieval precision/recall and answer relevance on a representative set of test queries.
5. Develop a working prototype of the AI assistant if time permits that answers natural-

language queries over a user's WhatsApp history, with retrieval processing kept on-device to preserve privacy .

4. Dataset / Data Source (if applicable)

- Mention potential datasets or data sources.
- Specify whether the data is public, proprietary, or to be collected.
- Provide links if available.
- If the dataset is not yet identified, mention the planned source.

Dataset / Data Source:

The main data source will be our own exported WhatsApp chat history consisting both of our personal and group chats that can be obtained from the "Export Chat" feature of WhatsApp. This provides us with real, messy and unstructured data to work with . The data will comprise of informal and formal chats, multiple speakers and fragmented data that show up in everyday chat rather than something cleaned or artificial

We decided to also look into a few of the publicly available NLP datasets that were used in building things for sentiment analysis so we may need to change them according to our needs Any personal data that we will be using will be anonymized where ever needed and we will be careful on how to handle it as we are aware how sensitive chat history can be.

5. Expected Deliverables

- Mention the expected outcomes of the project.
- Focus on tangible outputs.
- Examples: software prototype, mobile application, web application, AI model, dashboard, dataset, research paper, etc.

Expected Deliverables:

1. A comparative study report that analyze the existing RAG frameworks (LangChain, LlamaIndex, Haystack) and memory architectures and document their specific limitations when applied to WhatsApp-style conversation data.
2. A custom retrieval pipeline which incorporates thread-aware chunking, speaker-aware context, and time-weighted ranking, adapted for multi-party, informal, code-mixed conversational data.
3. Evaluation benchmark, including a small labeled set of natural-language queries and ground-truth answers derived from real chat data, used to measure retrieval and answer accuracy.
4. Working prototype of the AI assistant that lets a user query their own WhatsApp chat history in natural language and get accurate, relevant answers .

6. Preliminary Literature Survey

- Identify at least five (not limited) relevant recent-year research papers.
- Prefer papers from reputed journals/conferences.
- Published survey papers need not be taken as your base papers but can be considered for reference purposes.
- Briefly summarize the contribution of each paper in one or two sentences.

No.	Paper Title	Year	Key Contribution
1	SECOM: Towards Effective Memory Granularity for Retrieval-Augmented Response Generation in Long-Term Conversations (Pang et al., ICLR)	2025	Shows that turn-level and session-level memory granularities are suboptimal for long-term conversations; proposes segment-level memory with compression-based denoising to improve retrieval.
2	MemGPT: Towards LLMs as Operating Systems (Packer et al.)	2023	Proposes an OS-inspired memory management system that pages context between short-term and long-term memory tiers, forming a foundational architecture for memory-augmented conversational agents.
3	Toward Conversational Agents with Context and Time Sensitive Long-Term Memory (Alonso et al.)	2024	Introduces methods for incorporating temporal context into memory retrieval, directly addressing time-dependent relevance in conversational data.
4	From RAG to Memory: Non-Parametric Continual Learning for Large Language Models (Gutiérrez et al.)	2025	Explores transitioning from standard retrieval-augmented generation to memory-based continual learning, bridging RAG frameworks and memory architectures.
5	MultiHop-RAG: Benchmarking Retrieval-Augmented Generation for Multi-Hop Queries (Tang and Yang)	2024	Introduces a benchmark for evaluating RAG systems on queries requiring reasoning across multiple retrieved pieces of context, offering a reference methodology for evaluation design.
6	Zep: A Temporal Knowledge Graph Architecture for Agent Memory (Rasmussen et al.)	2025	Represents agent memory as a temporal knowledge graph to capture time-evolving relationships between facts, relevant to designing time-weighted retrieval.
7	Evaluating Very Long-Term Conversational Memory of LLM Agents (Maharana et al.)	2024	Proposes an evaluation methodology for measuring how well LLM agents retain and retrieve information across extended, long-term conversation histories.

6.1 Comparison of Related Work

- Compare the five (not limited) shortlisted papers side-by-side using consistent aspects.
- Suggested aspects: methodology/technique used, dataset, key contribution, and limitations/research gap — adapt columns to fit your domain.
- This table should make the research gap motivating your project immediately visible.

Table 1: Comparison of related work across key aspects

No.	Paper / Ref.	Year	Methodology	Dataset Used	Limitations / Research Gap
1	Pang et al. (SECOM)	2025	Segment-level memory construction with compression-based denoising for retrieval-augmented response generation	Long-term conversational dialogue benchmarks (multi-session, single-agent-user dialogue)	Designed for single-user AI-assistant dialogue sessions; does not address multi-party human conversations, code-mixed language, or fragmented short messages.
2	Packer et al. (MemGPT)	2023	OS-inspired memory paging between short-term (in-context) and long-term (external) memory tiers	Synthetic and conversational agent dialogue tasks	Memory management assumes a single ongoing dialogue between user and agent; no handling of multiple human speakers, informal/code-mixed text, or exported chat log formats.
3	Alonso et al.	2024	Time-sensitive long-term memory architecture incorporating recency and temporal context into retrieval	Simulated long-term conversational agent dialogues	Focuses on temporal relevance within single-user assistant dialogue; does not address multi-party speaker attribution or informal chat structure.
4	Gutiérrez et al.	2025	Non-parametric continual learning approach bridging RAG and memory-based updating	General-purpose knowledge/QA corpora	Focuses on factual knowledge continual learning rather than personal, conversational, multi-speaker data; privacy is not a primary consideration.
5	Tang and Yang (MultiHop-RAG)	2024	Benchmark construction for evaluating RAG retrieval on multi-hop reasoning queries	Curated multi-hop QA dataset built from structured/news articles	Evaluation benchmark built on structured documents rather than conversational data; no benchmark exists for personal chat-based retrieval.

Note: Replace the placeholder entries (author, year, methodology, dataset, limitations) with details drawn from the five papers listed in Section 6. The `\cite{}` commands are optional — remove them if you are not using a BibTeX/.bib references file, or link them to your bibliography keys if you are.

6.2 Novelty / Research Gap Statement

- State, in 4–6 sentences, what specifically is missing or under-addressed across the five compared papers (refer back to the **Limitations / Research Gap** column above).
- Clearly state how the proposed project addresses this gap — avoid vague claims like “better accuracy”; be specific about the mechanism, data, or approach that is new.
- This statement should logically follow from Table 1 and directly justify the Objectives in Section 3.

Novelty / Research Gap:

On comparing the above papers a consistent gap can be seen ; these existing memory augmented and RAG based approaches are either for a single user , a AI assistant dialogue (like SECOM, MemGPT) or for structured document or knowledge text (Gutiérrez et al., MultiHop-RAG) and none of them address the conversation data that is exchanged between multiple users. None of them can properly handle the specific structural like WhatsApp's short fragmented messages , multiple speakers in the same chat, informal and reply relationships that could carry meaning beyond the text. The time sensitivity is only addressed by one paper (Alonso et al.) but even there it was applied to single user memory, not to a multiple user scenario where the relevance can depend who said what over different chats. Also, additionally there are no evaluated benchmark in this set that used real personal data for privacy reasons of such given data.

We decided to address this by first benchmarking the existing frameworks and the memory architectures on WhatsApp chat data to specifically identify where their chunking and retrieval methods break down on unstructured and fragmented WhatsApp data. Based on these findings we aim to design a retrieval mechanism that especially makes use of speaker identity and reply-thread structure into the chunk consideration and when combined with a time weight ranking will adjust the scores based on the message recency compared to the query. This method targets the gaps which are the multi party context and time sensitivity of conversations.

7. Potential Publication Venues

- Identify at least **3 conferences** (at least one from India) and **3 journals** relevant to the proposed work.
- Select venues that are appropriate for the project scope.
- Mention only venues that regularly publish work in AI, ML, Data Science, Computer Vision, NLP, Cybersecurity, IoT, Healthcare AI, or Software Engineering.
- Briefly justify why the venue is relevant.

Possible Conferences

NeurIPS, ICML, ICLR, AAAI, IJCAI, KDD, ICDM, SDM, AISTATS, UAI, CVPR, ICCV, ECCV, WACV, BMVC, ACL, EMNLP, NAACL, COLING, ECAI, ICPR, ICIP, ICASSP, COMSNETS, CODS-COMAD, INDICON, ICACCS, ICCCI, IEEE ANTS, COMPUTE, ICACC, TENCON

Possible Journals

IEEE Access, Expert Systems with Applications, Knowledge-Based Systems, Applied Soft Computing, Engineering Applications of Artificial Intelligence, Information Sciences, Pattern Recognition, Neurocomputing, Artificial Intelligence Review, Machine Learning, Neural Computing and Applications, Journal of King Saud University – Computer and Information Sciences, Multimedia Tools and Applications, Computers & Security, Computer Networks, Future Generation Computer Systems, Computer Methods and Programs in Biomedicine, Artificial Intelligence in Medicine, Data & Knowledge Engineering, ACM Transactions on Intelligent Systems and Technology (TIST), ACM Computing Surveys, IEEE Transactions on Artificial Intelligence, IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Knowledge and Data Engineering (TKDE), IEEE Transactions on Dependable and Secure Computing (TDSC), Scientific Reports, PLOS ONE, Sensors, Electronics, Heliyon

Selected Venues & Justification:

Conferences:

- **CODS-COMAD** (India) – Regularly features applied data science, ML, and NLP work at a scope suited to student and early-stage research; a realistic venue for our comparative RAG evaluation and prototype.
- **ICACCS** – Covers applied AI/ML systems and software engineering, and accepts application-oriented contributions such as framework evaluations and system prototypes, matching our project’s scope.
- **COLING** – Focused on NLP and language technologies including retrieval and dialogue systems; an internationally recognized venue for applied RAG/conversational-retrieval work at a more attainable scope than top-tier ACL/EMNLP tracks.

Journals:

- **IEEE Access** – Broad AI/ML scope with a faster review cycle and openness to applied, systems-level contributions rather than requiring purely theoretical novelty.
- **Expert Systems with Applications** – Focuses on applied AI systems solving real-world problems, matching our framing of a practical retrieval system for personal chat data.
- **Journal of King Saud University – Computer and Information Sciences** – Publishes applied AI/NLP and data engineering research at a scope and acceptance bar appropriate for a well-executed student project with empirical evaluation.

8. References

- Use IEEE citation format.
 - Include all papers, datasets, reports, and websites referred to during project identification.
 - Provide DOI or URL whenever available.
- [1] C. Pang *et al.*, “SECOM: Towards Effective Memory Granularity for Retrieval-Augmented Response Generation in Long-Term Conversations,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2025/file/e56f394bbd4f0ec81393d767caa5a31b-Paper-Conf.pdf
 - [2] C. Packer, V. Fang, S. G. Patil, K. Lin, S. Wooders, and J. E. Gonzalez, “MemGPT: Towards LLMs as Operating Systems,” *arXiv preprint arXiv:2310.08560*, 2023. [Online]. Available: <https://arxiv.org/abs/2310.08560>
 - [3] N. Alonso, T. Figliolia, A. Ndirango, and B. Millidge, “Toward Conversational Agents with Context and Time-Sensitive Long-Term Memory,” *arXiv preprint arXiv:2406.00057*, 2024. [Online]. Available: <https://arxiv.org/abs/2406.00057>
 - [4] B. J. Gutiérrez, Y. Shu, W. Qi, S. Zhou, and Y. Su, “From RAG to Memory: Non-Parametric Continual Learning for Large Language Models,” *arXiv preprint arXiv:2502.14802*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.14802>
 - [5] Y. Tang and Y. Yang, “MultiHop-RAG: Benchmarking Retrieval-Augmented Generation for Multi-Hop Queries,” *arXiv preprint arXiv:2401.15391*, 2024. [Online]. Available: <https://arxiv.org/abs/2401.15391>

- [6] Infobip, “WhatsApp Statistics 2026: Global Usage & Market Overview,” Infobip Blog, 2026. [Online]. Available: <https://www.infobip.com/blog/whatsapp-statistics>
- [7] Meta Platforms, Inc., “How to Export Your Chat History,” WhatsApp Help Center. [Online]. Available: <https://faq.whatsapp.com/1180414079177245>

8. Faculty Remarks

As guide, I hereby confirm that I checked the plagiarism and AI content.

Note: The submitted document must have a similarity index of less than 15% and an AI-generated content score of less than 20%, as verified using Turnitin.

Signature with Remarks by Guide

Signature of Project Coordinator

Name of Guide: _____

Name of Project Coordinator: _____